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Statistical Models
Week 5 : general results and methods for glm

How do we estimate parameters in a generalized linear models? How do we test hypotheses about the parameters?

You will fit glms to data, using R's function `glm`. You will learn about the theoretical background, maximum likelihood estimation in glms. We will also have a look at the asymptotic properties of MLE in the context of glm. If time allows, we discuss numerical techniques behind solving the maximum likelihood equations, in particular iterative reweighted least squares. The second lecture deals with model assessment. Goodness of fit of glm can be assessed by the deviance – a likelihood ratio statistic ( ) – or by studying residuals. We will also see how to estimate the dispersion parameter in situations where it is not given.

Relevant Readings

Davison, 10.2, 10.3.2, evtl. 10.4; supplemented by Agresti, 4.4.6, 4.5.

Lecture Material

- Link to onenotes and cheat sheet on Google drive
- "My first glm" Try fitting a binomial proportion glm on the butterfly data
- Analysis of a glm Residuals and model reduction in R (may be updated before Friday lecture)

Exercises 5

Problem 5.1: GLM for comparing two groups

(Compare Agresti 2015, Exercises 4.6 and 4.7)
In this exercise we consider data from two groups A and B of size n_A and n_B, that can be modelled by i.i.d. Poisson variables within each group (thus we have in all a sample of size $n = n_A + n_B$). The first n_A observations belong to a group A, and the remaining n_B belong to a group B, with different group means μ_A and μ_B. For the analysis by a GLM, we use the linear predictor $g(\mu_i) = \eta_i = \beta_0 + \beta_1 \mathbf{1}(i \in \{1, ..., n_A\})$.

a. Show that for the log-link function, the GLM likelihood equations imply that the fitted means $\hat{\mu}_A$ and $\hat{\mu}_B$ equal the sample means.

b. Show that the result of a) holds no matter what link function g is used or what exponential dispersion family distribution for the Y_i is assumed.

You could also try to generalize the result to more than two groups.

Problem 5.2: Survival times

Times to event (survival times) are often modeled by an exponential distribution. Recall that this is a special case of Gamma distribution :-)
The following data example deals with survival data of leukemia patients and is a classic [a]. Here y_i

represent times to death in weeks from diagnosis, and x_i the $\log_{10}$ of initial white blood cell count, for seventeen patients.
The R code below generates vectors `y` and `x`:

```
y <- c(65, 156, 100, 134, 16, 108, 121, 4, 39, 143, 56, 26, 22, 1, 1, 5, 65)
x <- c(3.36, 2.88, 3.63, 3.41, 3.78, 4.02, 4.00, 4.23, 3.73, 3.85, 3.97, 4.51,
       4.54, 5.00, 5.00, 4.72, 5.00)
```

a. Plot y_i against x_i. Do the data show any trend?

b. A possible specification for $\mathrm{E}\, y_i$ is

$$\mathrm{E}\,(y_i) = \exp(\beta_1 + \beta_2 x_i),$$

which will ensure that $\mathrm{E}\,(y)$ is non-negative for all values of the parameters and all values of x. Which link function is appropriate in this case?

c. Using R's function `glm`, fit a model that assumes that the y_i have an exponential distribution, with mean $\mathrm{E}\,(y_i)$ given in (b). Hint: use the Gamma family. The Gamma shape parameter has no influence on the estimate of the model parameters in a glm.

d. For the fitted model, compare the observed values y_i and fitted values $\hat{y}_i = \exp(\hat{\beta}_1 + \hat{\beta}_2 x_i)$, and use the standardized residuals $r_i = (y_i - \hat{y}_i)/\hat{y}_i$ to investigate the adequacy of the model.
(Note: $\hat{y}_i$ is used as the denominator of r_i because it is an estimate of the standard deviation of y_i— why is this?)

[a] data from the book Cox & Snell (1981) Applied statistics, found in Dobson & Barnett (2008): An Introduction to Generalized Linear Models

Problem 5.3: O-ring failure in space shuttles: glm and model checking

Davison, Practicals 10.1.
The `shuttle` data are unfortunately slightly corrupted – they have row names that are numbers, but they should be character.
Repair them by running

```
row.names(shuttle) <- as.character(row.names(shuttle))
```

or simply remove them with `row.names(shuttle) <- NULL` before doing any analysis.

Problem 5.4: Deviance goodness of fit test for the Poisson model - when does asymptotic theory work, and when not?

In the lectures, we have learned that the scaled deviance can be used to test if a model describes the data well, since scaled deviance between a model and the corresponding saturated model often has an asymptotic χ^2-distribution. In this exercise, you explore this for a Poisson model by simulation. Let $Y_1, \ldots, Y_n$ be independent Poisson distributed variables with $\mathrm{E}\, Y_i = \lambda_i$. This is the saturated model M_0. In the submodel M_1, all mean values coincide, that is, $\lambda_1 = \cdots = \lambda_n$.

a. Show that the scaled deviance for model M_1 is given as

$$D = 2\sum_{j=1}^{n} Y_j \log\left(\frac{Y_j}{\bar{Y}}\right), \quad \bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i,$$

where we set $Y_j \log(Y_j/\bar{Y}) = 0$ if $Y_j = 0$.

Extra (ad libitum): show that $\lim_{x\to 0} x\log(x) = 0$.

b. Simulate $N = 1000$ values of D and compare with a χ^2-distribution with $n-1$ degrees of freedom. For a visual comparison, you can for example use an appropriate qq-plot (not qqnorm). Also estimate the probability that D exceeds the 95%-quantile of the χ^2_{n-1}-distribution.

Do this for several values of λ and n, e.g. $\lambda = 1, 10, 100$ and $n = 10, 100, 1000$. In some cases, you will observe that the χ^2-approximation is not good. Summarize your findings - when would you recommend a goodness-of-fit test based on the χ^2-approximation of the scaled deviance, and when not?

Try to explain your findings.